

Computer Science

Yao Zhang

1	Deep Learning	1
2	Robot Learning	2
3	Causal Data Science	3
4	Concise Machine Learning	4
5	Artificial Intelligence Agents	6
6	Physics Aware Deep Learning	7
7	Mathematics of Deep Learning	8
8	Generative Artificial Intelligence	9
9	MultiModal Artificial Intelligence	10
10	Group Equivariant Deep Learning	11
11	Optimization for Artificial Intelligence	12
12	Machine Learning and Reinforcement Learning	14
13	Artificial Intelligence in the Sciences and Engineering	15
14	Biologically Inspired Artificial Intelligence and Optimization	16
15	High Dimensional Analysis: Random Matrices and Machine Learning	17
16	Mathematics of Massive Data Analysis: Fundamentals and Applications	19

1 Deep Learning

2 Robot Learning

Main Lectures

1. [Introduction to Robot Learning](#)
2. [Robot Control and MDPs](#)
3. [Imitation Learning](#)
4. [Reinforcement Learning 1](#)
5. [Reinforcement Learning 2](#)
6. [Generative Models](#)
7. [Sequence Modeling and Transformers](#)
8. [World Models](#)
9. [Generalist Robot Policies](#)
10. [Embodied Reasoning and Test-time Scaling](#)
11. [Frontier and Open Problem](#)

Guest Lectures

1. [Abishek Gupta](#)
2. [Danfei Xu](#)
3. [Aviral Kumar](#)
4. [Andrew Wagenmaker](#)
5. [Cheng Chi](#)
6. [Ted Xiao](#)
7. [Scott Reed](#)
8. [Quan Vuong](#)
9. [Archit Sharma](#)
10. [Lucas Beyer](#)

3 Causal Data Science

1. [Introduction](#)
2. [Markov Equivalence Class and D-Separation](#)
3. [Markov Equivalence Class and Ancestral Graphs](#)
4. [Partial Ancestral Graph and Fast Causal Inference Algorithm](#)
5. [Conditional Independence Tests & Possible D-Separating Sets](#)
6. [Midterm Review](#)
7. [Effect Identification](#)
8. [Adjustment Criterion for PAGs](#)
9. [Do Calculus](#)
10. [ID Algorithm](#)
11. [ID Algorithm for PAGs](#)
12. [Counterfactuals and Effect Estimation](#)

4 Concise Machine Learning

1. Introduction; Classification, Data Splitting, and Overfitting versus Underfitting
2. Linear Classifiers, Decision Functions, Boundaries, Centroid Method, and Perceptrons
3. Gradient Descent, Stochastic Gradient Descent, Perceptron Learning, Feature vs Weight Space, Hard-Margin SVM
4. Soft-Margin SVM, Features, and Nonlinear Decision Boundaries
5. Machine Learning Abstractions: Application, Data, Model, Optimization Problem, and Algorithm
6. Decision Theory, Bayes Rule and Risk, Generative and Discriminative Models
7. Gaussian Discriminant Analysis, QDA, LDA, Maximum Likelihood Estimation, Isotropic Gaussian Fitting
8. Eigenvalues, Eigenvectors, Eigendecomposition, Quadratic Forms, Ellipsoids, Covariance, Multivariate Normal Distributions
9. MLE, QDA, and LDA Revisited for Anisotropic Gaussian Distributions
10. Regression, Least Squares, Design Matrix, Normal Equations, Pseudoinverse, Logistic Regression, Gradient Descent
11. Newton's Method, Logistic Regression, LDA Comparison, ROC Curves, Weighted and Polynomial Regression
12. Regression Justification, Empirical Risk, Maximum Likelihood, Bias-Variance Tradeoff, Underfitting and Overfitting
13. Ridge Regression, MAP Regularization, Subset Selection, and Lasso for Reduced Overfitting
14. Decision Trees, Tree-Building Algorithms, Entropy, and Information Gain
15. Decision Tree Regression, Pruning, Multivariate Splits, Ensemble Learning, Bagging, and Random Forests
16. Neural Networks, Gradient Descent, and Backpropagation Algorithm
17. Vanishing Gradients, ReLUs, Softmax Backpropagation, Cross-Entropy Loss, and Computational Model Differences
18. Neuron Biology, Axons, Dendrites, Synapses, Action Potentials, and Faster Training Heuristics

19. Convolutional Neural Networks, Retinal Ganglion Cells, and Simple and Complex Visual Cortex Cells
20. Unsupervised Learning, PCA, Maximum Likelihood, Variance Maximization, Projection Error Minimization, Eigenfaces
21. SVD for PCA, K-Means, Hierarchical Clustering, and Greedy Agglomerative Clustering
22. Pseudoinverse, SVD, Neural Network Generalization, Data Augmentation, L2 Regularization, Dropout, Double Descent
23. Batch Normalization, ResNets, and Accelerated Optimization with AdamW
24. AdaBoost, Nearest Neighbor Classification, and Its Relation to Bayes Risk
25. K-Nearest Neighbor Search, Query Acceleration, Voronoi Diagrams, Point Location, and K-D Trees

5 Artificial Intelligence Agents

1. [What are Agents and How Do They Work?](#)
2. [Tool Use for Language Model Agents](#)
3. [Long Context Modeling for Agents](#)
4. [Memory and Skills for Agents](#)
5. [text](#)
6. [text](#)

6 Physics Aware Deep Learning

1. [Introduction, Types of Machine Learning Problems in Physical Sciences](#)
2. [Multi-layer Perceptrons and Deep Neural Networks](#)
3. [Stochastic Gradient Descent; Backpropagation, Bias-Variance Trade-Off; Regularization](#)
4. [Introduction to PyTorch](#)
5. [Convolutional Neural Networks](#)
6. [Structural Health Monitoring with CNN in PyTorch](#)
7. [How to Train CNNs](#)
8. [CNNs Implementation](#)
9. [Physics-Informed Neural Networks](#)
10. [Introduction to PINN in PyTorch: Harmonic Oscillator](#)
11. [Physics-Informed Neural Networks Continued](#)
12. [More on PINNs: Heat Equation](#)
13. [Neural Operators](#)
14. [Fourier Neural Operators in PyTorch](#)
15. [Physics-Aware Recurrent Convolutions](#)
16. [Introduction to PARC in PyTorch](#)
17. [Deep Reduced Order Modelling](#)
18. [Autoencoders in PyTorch and Latent Dynamics Learning](#)
19. [Generative Neural Networks](#)

7 Mathematics of Deep Learning

1. The Curse of Dimensionality in Machine Learning
2. Function Approximation. Linear vs Non-Linear Approximation
3. Universal Approximation Theorem for Shallow Nets. Harmonic Analysis for Shallow Nets
4. Reproducing Kernel Hilbert Spaces
5. Learning in the Overparametrised Regime. Generalisation Bounds for Normed spaces
6. Convex Optimization Reminders
7. Optimization in Shallow Spaces: the linear regime. The Neural Tangent Kernel
8. Nonlinear overparametrised regime. Mean-Field description
9. Lower bounds: Statistical-Queries, Reductions
10. Depth Separation
11. Invariances and Symmetries
12. Implicit Regularisation. Noisy dynamics. Gradient Flow vs Gradient Descent vs Stochastic Gradient Descent
13. Generative Modeling: Diffusion

8 Generative Artificial Intelligence

1. [Introduction to Machine Learning](#)
2. [Back Propagation and Training Process](#)
3. [Back Propagation and Cost Function](#)
4. [Loss Function of Generative AI](#)
5. [Generative AI Architecture & Entropy Loss Function](#)
6. [Overall Perspectives of Generative AI Architecture](#)
7. [Sequential generative AI with RNN](#)
8. [Back Propagations in RNN](#)
9. [Attention is All You Need](#)
10. [Multi-Head Attentions in Transformer Encoder](#)
11. [Cross-Attentions in Transformer Encoder-Decoder](#)
12. [Generative AI for Image, Generative Adversarial Network Architecture & Loss Functions](#)
13. [Variational Autoencoder Architecture](#)
14. [Variational Autoencoder Loss Function and Image Generation](#)
15. [Diffusion Model for Image Generation](#)
16. [Reinforcement Learning for Fine Tuning and Human Feedback](#)
17. [Markov Property and Bellman Equation](#)
18. [Monte Carlo Method & Temporal Difference Method](#)
19. [Deep Reinforcement Learning for Value Agent Network](#)
20. [Deep Reinforcement Learning for Policy Agent Network](#)

9 MultiModal Artificial Intelligence

1. [Introduction](#)
2. [Multimodal Research Tasks](#)
3. [Data and Heterogeneity](#)
4. [Multimodal Fusion](#)
5. [Multimodal Alignment](#)
6. [Large Multimodal Models](#)
7. [Multimodal Generation](#)
8. [Modern Generative AI](#)
9. [Multimodal Reasoning](#)
10. [Multimodal Interaction](#)
11. [Cross-Modal Transfer](#)
12. [Self-Evolving AI](#)
13. [Human-AI Interaction](#)

10 Group Equivariant Deep Learning

1. Regular group convolutional neural networks

- (a) [Introduction](#)
- (b) [Group theory: Groups, group product, inverse, action, representation, affine groups \$G = \mathbb{R}^d \rtimes H\$](#)
- (c) [Regular group convolutions: Template matching viewpoint](#)
- (d) [SE\(2\) Equivariant NN Example: histopathology](#)
- (e) [A brief history of G-CNNs](#)
- (f) [Group Theory: Transitive action, homogeneous space, quotient space](#)
- (g) [Group convolutions are all you need! \(Equivariant linear layers between feature maps are group convolutions\)](#)

2. Steerable group convolutional neural networks

- (a) [Steerable kernels/basis functions \(Definition and SO\(2\) example\)](#)
- (b) [Revisiting regular G-convs with steerable kernels \(Motivating the Fourier transform\)](#)
- (c) [Group Theory: Irreducible representations and Fourier transform](#)
- (d) [Group Theory: Induced representations and feature fields](#)
- (e) [Steerable group convolutions, and how to use them](#)
- (f) [Activation functions for steerable G-CNNs](#)
- (g) [Derivation of Harmonic networks from regular g-convs](#)

3. Equivariant graph neural networks

- (a) [Motivation for SE\(3\) equivariant graph NNs](#)
- (b) [Equivariant message passing as non-linear convolution](#)
- (c) [Tensor products as conditional linear layers \(and MLPs\)](#)
- (d) [Group Theory: SO\(3\) irreps \(Wigner-D matrices\), Clebsch-Gordan TP](#)
- (e) [Literature: 3D Steerable \(graph\) convolutions](#)
- (f) [Literature: Regular \(as opposed to steerable\) equivariant graph NNs](#)
- (g) [Literature: Gauge equivariant graph NNs](#)

11 Optimization for Artificial Intelligence

Optimization for Machine Learning

1. [Introduction and Course Overview & Convex Sets and Convex Functions](#)
2. [Convex Functions Continued & Gradient Descent and Family](#)
3. [Gradient Descent Continued & Projected and Proximal Gradient Descent](#)
4. [Projected Gradient Descent and Conditional Gradient Descent & Newton, Quasi Newton and Second Order Methods](#)
5. [Second Order Methods Cont., Barzelia Borwein GD and Conjugate Gradient Method & Coordinate Descent Algorithms](#)
6. [Stochastic Gradient Descent and Family & Variants of SGD for Deep Learning](#) [A](#) [B](#)
7. [Submodular Functions: Definitions, Examples, and Information Measures](#) [A](#) [B](#)
8. [Submodular Functions: Definitions, Examples, and Information Measures](#)
9. [Submodular Function Optimization](#)
10. [Submodular Function Optimization](#)
11. [Applications of Continuous and Discrete Optimization](#)

Nonconvex Optimization for Deep Learning

1. Linear Regression and GD on Quadratics [A](#) [B](#)
2. Neural Networks Basics, Stationary Points and Noisy Gradients [A](#) [B](#) [C](#)
3. Saddle Points and Initialization of Deep MLPs [A](#) [B](#)
4. Non-convex Convergence Rates [A](#) [B](#) [C](#)
5. SGD Complexity and Learning Rate Scheduling [A](#) [B](#) [C](#)
6. Generalization and Local Minima [A](#) [B](#) [C](#)
7. Neural Tangent Kernel [A](#) [B](#) [C](#)
8. μ P, Hyperparameter Transfer and Overparametrization [A](#) [B](#) [C](#)
9. Training Deep Networks: Skip Connections and Batch Normalization [A](#) [B](#) [C](#)
10. Transformers, Layer Norm and Rank Collapse [A](#) [B](#) [C](#)
11. Adaptive Optimization Methods: Adam and Friends [A](#) [B](#) [C](#)
12. More Adam and Training LLMs [A](#) [B](#)
13. Bonus [A](#)

12 Machine Learning and Reinforcement Learning

1. [Introduction to Machine Learning and Reinforcement Learning](#)
2. [Perceptron and Forward Propagation](#)
3. [Back Propagation](#) [A](#) [B](#)
4. [Entropy and Cost Function](#) [A](#) [B](#)
5. [Softmax and Regressions](#)
6. [Machine Learning Design and Performance](#)
7. [Overview of Deep Reinforcement Learning and Alphago](#)
8. [Recurrent Neural Network Architecture](#)
9. [Recurrent Neural Network and Long Short-Term Memory \(LSTM\)](#)
10. [RNN Back Propagation Through Time](#)
11. [LSTM Back propagation and Transformer Learning](#)
12. [CNN Architectures](#)
13. [Autoencoder and GAN](#)
14. [Markov Decision Process and Bellman Equation](#)
15. [Model based Prediction and Control](#)
16. [Monte Carlo Method and Temporal Difference Method](#)
17. [Value Based DNN Agent](#)
18. [Policy Based DNN Agent](#)

13 Artificial Intelligence in the Sciences and Engineering

1. [Course Introduction](#)
2. [Introduction to Deep Learning](#)
3. [Physics-Informed Neural Networks](#)
 - (a) [Introduction](#)
 - (b) [Theoretical Insights](#)
4. [Operator Learning](#)
 - (a) [Introduction](#)
 - (b) [FNO](#)
- (c) [ReNO](#)
- (d) [Operator Transformer](#)
- (e) [Graph-based Model](#)
- (f) [GAOT](#)
5. [Generative model for PDE: GenCFD](#)
6. [Foundation models of PDEs: Poseidon](#)
7. [Applications of AI in Chemistry Biology: A B](#)

14 Biologically Inspired Artificial Intelligence and Optimization

1.
 - (a) Introduction to the Course & Its Policies & Its Motivation
 - (b) Evolutionary Approach to Engineering Design Optimization
 - (c) Population Genetics of Evolutionary Algorithms
 - (d) The Four Forces of Evolution and The Drift Barrier
 - (e) The Basic Genetic Algorithm and Its Implementation
 - (f) GA Wrap: Options for Selection & Crossover & Mutation & Extensions
2.
 - (a) Evolutionary Computing from Optimization to Programming
 - (b) Genetic Programming, Immunocomputing & Artificial Immune Systems
 - (c) Immunocomputing: Genetic Approaches for Diverse Solution Portfolio
3.
 - (a) Multi-Criteria Decision Making, Pareto Optimality & Intro to MOEA
 - (b) Multi-Objective Genetic Algorithms: Weight/Vector-Based Approaches
 - (c) Pareto Ranking and Moving from Communities to Meta-Populations
4.
 - (a) From DGA/PGA to Niching Methods for Multi-Modal Optimization
 - (b) Niching Methods in Multi-Modal Optimization
5.
 - (a) Introduction to Simulated Annealing and Entropy
 - (b) From Maximum Entropy Toward Optimization by Sim Annealing
 - (c) Toward SA: Intro to Boltzmann Sampling and Monte Carlo Integration
 - (d) Simulated Anneal & Wrap and Introduction to Ant Colony Opt
6.
 - (a) Ant Colony Opt. and Intro. to Bacterial Foraging Opt.
 - (b) Bacterial Foraging Opt. & Particle Swarm Optimization
7.
 - (a) Neural Foundations of Learning
 - (b) Feeding Forward from Neurons to Networks
 - (c) Recurrent Networks and Temporal Supervision
 - (d) Reinforcement Learning: Active Learning in Rewarding Environments
 - (e) Learning without Teacher: Unsupervised & Self-Supervised Learning
 - (f) Spiking Neural Networks and Neuromorphic Computation
8.
 - (a) Complex Systems Models of Computation: Cell. Automata & Neighbors

15 High Dimensional Analysis: Random Matrices and Machine Learning

1. Introduction and Survey
2. Volumes in High Dimensions
3. Concentration of Volumes
4. Gaussian Random Vectors and Concentration of Their Norm
5. Exponential Concentration of Norm of Gaussian Random Vectors
6. Non-Linear Concentration of Gaussian Random Vectors for Lipschitz Functions
7. Proof of Non-Linear Concentration for Gaussian Random Vectors
8. General Remarks on Linear and Non-Linear Concentration Inequalities
9. Wishart Random Matrices and Concentration of Largest Eigenvalue
10. Proof of Concentration of Largest Eigenvalue
11. The Marchenko-Pastur Law for Wishart Matrices
12. Preparations for Proof of Marchenko-Pastur Law
13. Proof of Marchenko-Pastur: Equation for Stieltjes Transform
14. Proof of Marchenko-Pastur: Stieltjes Inversion Formula
15. Spiked Signal-Plus-Noise Model
16. Proof of the Signal-Plus-Noise Theorem
17. Double Descent and Linear Regression: Over-Determined Case
18. Double Descent and Linear Regression: Under-Determined Case
19. General Remarks on Random Feature Model
20. Asymptotic Eigenvalue Distribution in the Random Feature Model
21. Another Proof of Marchenko-Pastur – Via Stein's Identity
22. Resolvent Method and Cumulant Expansion
23. Cumulants and Their Properties and Uses
24. Calculation of the Random Feature Eigenvalue Distribution

25. [Gaussian Equivalence Principle for Non-Linear Random Features](#)
26. [Gradient Descent for Linear Regression](#)
27. [Time Evolution of Learning and Neural Tangent Kernel](#)
28. [Properties of the Neural Tangent Kernel](#)
29. [Free Probability Theory and Linearization of Non-Linear Problems](#)
30. [The Curse and the Blessing of High Dimensions](#)

16 Mathematics of Massive Data Analysis: Fundamentals and Applications

1. Concentration Inequality; Geometry of High-Dimensional Euclidean Space
2. Hyperspheres and Hyperballs are Weird in High Dimensions
3. Gaussian Annulus Theorem and Johnson-Lindenstrauss Lemma
4. Markov Chains and MCMC
5. Convergence of Random Walks on Undirected Graphs Using Conductance
6. More Markov Chains
7. (a) Volume via MCMC and PageRank
(b) Overview
8. Overview of Hashing, Sketching, and Random Graph Theory
9. (a) Overview of Wavelets, Low-Dimensional Embeddings, and Persistent Homology
(b) Hashing Theory
10. K-Independent Hash Functions
11. Frequency Moments of Data Streams and an Idealized Count-Distinct Sketch
12. Non-Idealized Count-Distinct Sketch and F2 Sketch
13. Frequent Items Sketch, MinHash Sketch
14. Erdos-Renyi Random Graphs and Phase Transitions
15. Erdos-Renyi Diameter and Isolated Vertices Phase Transitions
16. Erdos-Renyi Component Size Phase Transitions
17. Percolation Theory
18. Finishing up Percolation Theory, and Intro to Wavelets
19. Finishing up Haar Wavelet
20. Necessary Conditions for General Systems of Wavelets
21. Sufficient Conditions for General Systems of Wavelets. The proof of lemma 11.6 is incorrect and revised as computation 11.6.

22. [The Fast Fourier Transform](#). Symmetry if real proof error; complex conjugates added.
23. [Matrix Sketching](#)
24. [VC-Dimension](#)
25. [Generalization Error in Classification](#)
26. [Representing Topological Spaces by Decomposing into Complexes](#)
27. [More Complexes and Introduction to Chains](#)
28. [Computing Simplicial Homology](#)
29. [Morse Functions](#)
30. [Morse-Smale Complex and Piece-Wise Linear Morse Functions](#)
31. [HyperMinHash](#)
32. [Secure Federated Count-Queries](#)